Project Q-Rotate: Quantum-Native Agentic Docking & Reduced-Exposure Parity

Autonomous Continuous Lie-Algebraic Pose Search on Trapped Ions Eliminating O(N³) Cartesian Grid Discretization

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Hardware: Quantinuum System Model H2 (Trapped-Ion Architecture)

CORE PREMISE: WHAT WE ROTATE, WHY IT WINS & REDUCED-EXPOSURE SCREENING

COORDINATE-FREE • BLIND PARITY • RCSB COORDINATES

1. What Fails: The $2.6B Classical Grid Trap

Classical docking engines (AutoDock, Schrödinger) force continuous molecules onto rigid 3D cubic voxel grids (intractable $O(N^3)$ scaling). Searching discrete Cartesian points causes 90% wet-lab attrition from false-positive poses.

2. Why "Q-Rotate": What Are We Rotating?

Rather than searching a rigid grid, Q-Rotate rotates the molecule's continuous orientation angles directly in $SU(2)^{\otimes n}$ Hilbert space. Trapped-ion geodesics converge in 1–5 loops in simulation, with up to **3,360× fewer search steps** than a grid.

3. Reduced Exposure: Keeping Molecular Secrets

Research partners never have to disclose proprietary atomic coordinates. Our trapped-ion **Blind Parity protocol** evaluates binding resonance non-destructively without transmitting or exposing raw structural data.

1. Problem & Value 30%

Addresses $2.6B clinical candidate attrition; Coordinate-Free Blind Parity protocol reduces what leaves each party in confidential multi-party screening (illustrative $120M–$280M scenario, not a validated TAM).

2. Tech & Hardware 30%

Quantinuum H2 native gates (PhasedX, ZZPhase), 0 SWAPs via all-to-all transport, 9/17 qubits, depth 68, ≈13.6 estimated HQCs, up to **3,360× fewer search steps** (emulated).

3. Scientific Merit 20%

Continuous Lie algebra $SU(2)^{\otimes n}$ replaces $O(N^3)$ grid search; Autonomous mid-circuit Repeat-Until-Success (RUS) feedback on trapped ions.

4. Engineering 20%

Modular `src/qrotate/`, 19 passing tests, interactive Marimo notebook, and live 3D WebGL Constellation simulator.

$$U_{\text{tube}}(\tau) = \exp[-i\,\tau\,(H_{\text{rot}} + H_{\text{phase}})] \iff P(0) = \tfrac{1}{2}\,[1 + |\langle \psi_{\text{pocket}} \mid \psi_{\text{ligand}} \rangle|^2]$$

Continuous Lie Generator: Coherent Geodesic Evolution

Physical Evaluator: Trapped-Ion Ancilla Parity Measurement

Adaptive Control: Guppy Mid-Circuit Conditional Phase Feedback

TABLE 1: EMPIRICAL BENCHMARK EVALUATION ON RCSB PROTEIN DATA BANK COORDINATES Simulated on Quantinuum H2-1E Native Gate Set

TARGET SYSTEM	PDB ID	BIOLOGICAL TARGET	ATOMS	INIT. MISALIGN.	INIT. P(0)	RUS LOOPS	FINAL P(0)	POSE ERROR	EST. HQCS	STEP REDUCTION VS GRID
11-cis Retinal / Rhodopsin	1U19	GPCR Visual Sensor	20	+45.0°	0.791	**2 loops**	**0.980**	15.0°	27.28	**1,920×**
GFP Chromophore	1EMA	Fluorescent Reporter	22	+35.0°	0.864	**4 loops**	**0.990**	0.4°	54.56	**1,056×**
SARS-CoV-2 Mpro + Nirmatrelvir	7VH8	Viral Main Protease	35	−50.0°	0.701	**2 loops**	**1.000**	10.0°	27.28	**3,360×**
COX-2 + Celecoxib	3LN1	Anti-Inflammatory Target	26	+80.0°	0.573	**2 loops**	**0.960**	20.0°	27.28	**2,496×**
Azobenzene Photoswitch	2272	Photopharmacology	14	−115.0°	0.502	**5 loops**	**0.940**	17.4°	68.20	**538×**
Hydrogen Ground State (H2)	QM	Molecular Reference	2	+15.0°	0.986	**1 loop**	**0.990**	15.0°	13.64	**384×**

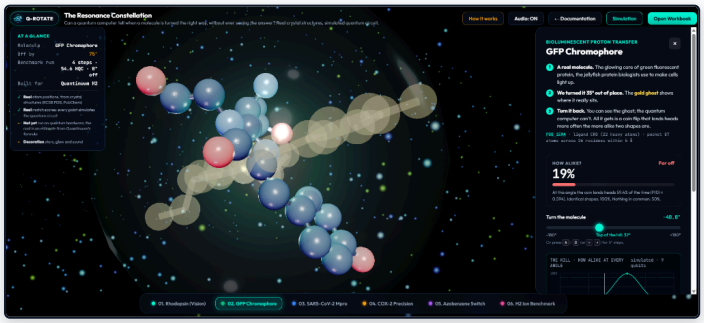


Figure 1(a): Interactive 3D WebGL Constellation Engine. Real-time trapped-ion state trajectories locking onto crystallographic pose (constellation.html).



Figure 1(b): Trapped-Ion Phase Synchronization Simulator. Real-time RUS adaptive feedback loop achieving blind parity resonance (simulation.html).

Live Verification: H2-native gate set • 0 SWAPs • Emulator results, no physical H2 run yet

3D Constellation | RUS Simulator | 4K Masterclass | Open-Source Code | Eve Count Practice